# Human-Computer Interaction

# Method-Design Crit

## Professor Bilge Mutlu

# Run-of-show (75 min)

- **0:00 · Frame:** today's method *is* today's crit lens
- **0:03 · Teaching (17 min):** measures · scales · power
- **0:20 · Set up:** tables, roles, the form, the timer
- **0:23 · Crit (42 min):** each team gets a full turn
- **1:05 · Debrief (10 min):** sharpest issues, patterns across tables

# Today's method *is* today's crit lens

A focused 17 minutes on **what makes a study design sound**, then you apply it as a reviewer on real designs: your classmates'.

*Bring: your Step-by-Step Experimental Design. That is the artifact you crit.*

# Teaching block: *measures • scales • power*

# (a) Choosing measures

Four kinds; match the **measure** to the **construct**:

1. **Objective:** against a standard (task success, time) · *Ch. 4–5*
2. **Behavioral:** observed actions (edits, accepting AI suggestions, reliance) · *Ch. 7*
3. **Subjective:** self-report (trust, satisfaction, ownership) · *Ch. 6*
4. **Physiological:** from the body (eye tracking, GSR, EEG) · *Ch. 7*

# Example measures: *Tullis & Albert*[1]

| Ch. | Metrics | Examples |
|---|---|---|
| 4 | Performance | task success, time on task, errors, efficiency, learnability |
| 5 | Issue-based | usability issues, severity ratings |
| 6 | Self-reported | rating scales, post-task and post-session ratings, SUS |
| 7 | Behavioral & physiological | verbal expressions, eye tracking, emotion, stress |
| 8 | Combined & comparative | single usability scores, scorecards, comparison to goals |

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[1] Tullis, T. & Albert, B. (2013). *Measuring the User Experience* (2nd ed.). Morgan Kaufmann. UW Library e-book

# Measures must be *valid*

Are you measuring what you claim?

- **Construct:** captures the intended concept (convergent + discriminant)
- **Criterion:** relates to or predicts an established measure
- **Content:** covers all relevant facets
- **Ecological:** generalizes to the real setting

*Most common crit finding: a measure that misses the construct.*

# (b) Scale development

A **scale** measures a construct through a set of **items**.

- **Construct:** the concept (trust, usability)
- **Item:** one question capturing part of it
- **Likert:** agreement with statements; combine items into one score
- Prefer an **existing validated scale** (SUS, UEQ, trust in automation[2]) over a home-grown one

[2] Jian, J.-Y., Bisantz, A. M. & Drury, C. G. (2000). Foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, 4(1), 53–71.

# Scales must be *reliable*

Consistency across items and repeats.

- **Cronbach's α:** internal consistency; **α ≥ .70** acceptable, **≥ .80** good
- Validity *and* reliability: a measure can be consistent yet wrong
- Building your own scale? Define the construct, generate items, factor-analyze, test α

# (c) Power & sample size

Can your design **detect** the effect it's looking for?

- **Effect size:** how big is the effect? (Cohen's *d*: .2 / .5 / .8)
- **α:** false-positive rate (conventionally .05)
- **Power (1 − β):** chance of detecting a real effect (target **.80**)
- Together they fix the **required sample size**

# From effect size to *n*

The pre-study move, not the post-hoc excuse:

1. Name the **smallest effect worth detecting** (from theory or prior work)
2. Set **α = .05**, **power = .80**
3. Compute the **required sample size**, e.g., in G*Power
4. Check it against your **feasibility**

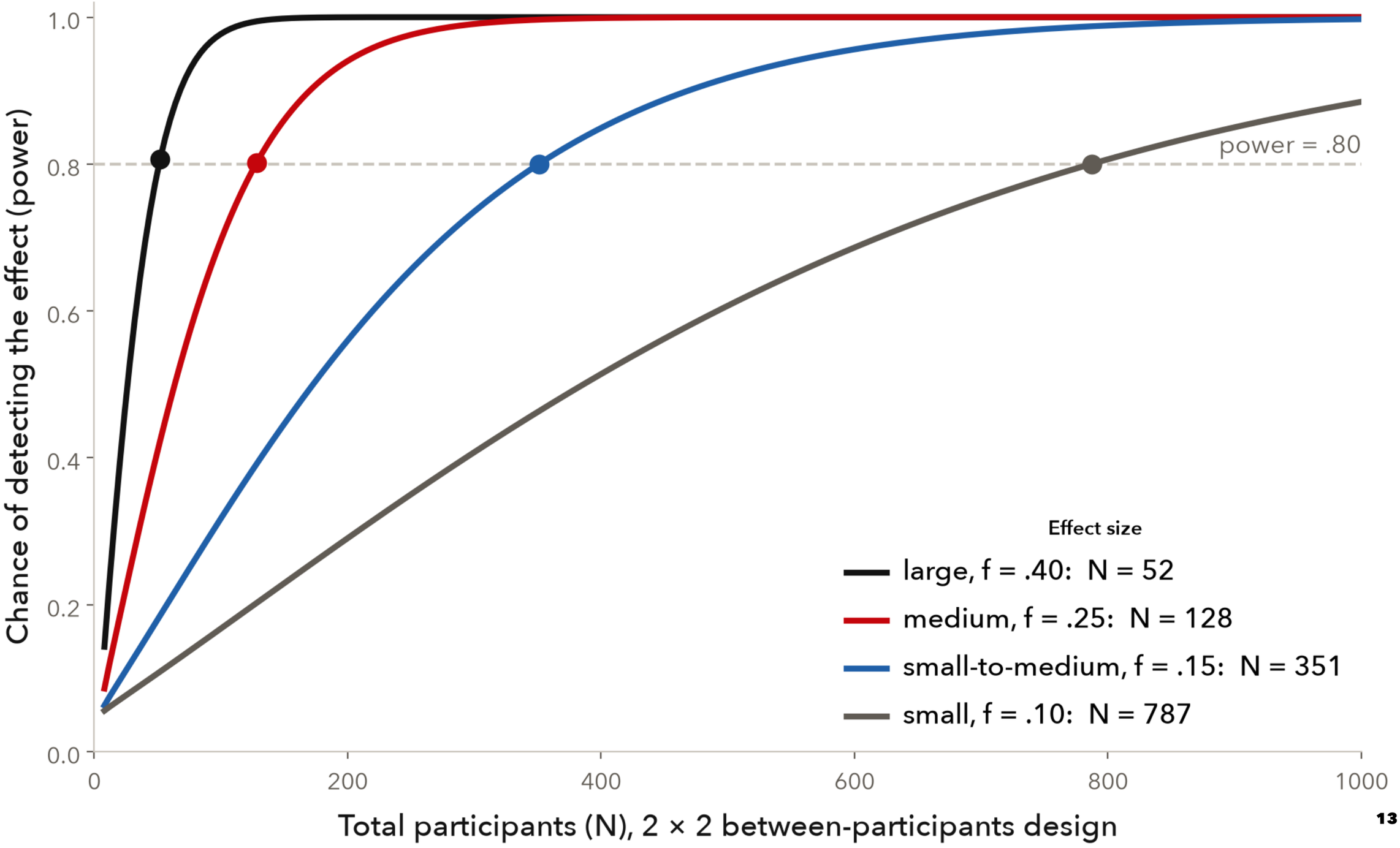
*Need more people than you can recruit? Fix the design now.*

# Worked example: *AI timing* × *experience*

2 × 2 between participants · α = .05 · power = .80

- **Timing effect**, medium (*f* = .25): **N = 128**, 32 per cell
- **Interaction**, smaller (*f* = .15): **N = 351**
- **Small effect** (*f* = .10): **N = 787**

Interactions are usually smaller than main effects, so they need far more people.


Chance of detecting the effect (power)
1.0
0.8
0.6
0.4
0.2
0.0
power = .80
Effect size
large, f = .40: N = 52
medium, f = .25: N = 128
small-to-medium, f = .15: N = 351
small, f = .10: N = 787
0
200
400
600
800
1000
Total participants (N), 2 × 2 between-participants design

# The research crit card

Today's teaching in four lines; it's on your crit form:

- **RQ ↔ method fit:** does the design answer the question?
- **Validity:** internal (confounds) · external (generalization) · construct (measuring what you claim)
- **Measures & power:** right measures for the construct; enough *n* to detect the effect
- **Feasibility:** doable in the time and resources left

# Set up the crit

# Tables, roles, timer

- **Your table is your crit group:** your team and the other team at it
- **A team on its own** joins a neighbouring table, making three
- **Roles** (rotate each crit):
  - **Facilitator:** runs the round-robin, makes sure everyone speaks, reports at debrief
  - **Timekeeper:** backstops the timer
  - **Scribe:** captures feedback on the form

# The per-team cycle

| Phase | 2-team table | 3-team table | What happens |
|---|---|---|---|
| **Present** | 6 min | 4 min | Design + the question you want crit on |
| **Silent writing** | 3 min | 2 min | Each reviewer writes on the form |
| **Round-robin** | 9 min | 6 min | One strength + one concern each |
| **Respond & commit** | 3 min | 2 min | Name **one change** you'll make |

Two turns of 21 min, or three of 14. Both end at the same time.

# Feedback etiquette

**Giving:** describe, don't judge ("this is hard to interpret because…") · ask questions ("what made you choose that design?") · build on ideas · be kind and specific · focus on the research goal, not preference.

**Receiving:** listen without defensiveness · ask follow-ups · it's about the work, not you.

# Capture it on the form

The written record is the deliverable, not feedback that evaporates.

- **One form per table:** File → Make a copy
- One section per presenting team, ending in its **committed revision**
- Each team submits its section on Canvas **by 11:59 pm tonight**

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# Debrief (10 min)

# Debrief

- **Facilitators** surface the sharpest issue their table saw
- Patterns across tables, tied back to the teaching block
- Usual suspects: *underpowered designs · measures that don't match the construct · RQ ↔ method mismatch*

# What's Next?

- **Tonight, 11:59 pm:** your team's crit section + committed revision, on Canvas
- **Mon Oct 19 · Ubiquitous, Mobile & Tangible:** read Weiser (1991), Dourish (2004); post your provocation by 9:30
- **Fri Oct 23 · Qualitative Analysis:** read Muller (2014), Xiao et al. (2023)
- **Project:** finalize your method by ~Oct 31